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% Econ 527 Spring 2026  
% HW 4  
% Professor Greaney  
% Written by: Camille Bergeron  
% Due: May 22, 2026

## Question 1: Hugget Model

```
run('programs/PS4_hugget.m'); % run Hugget
```

Transition probabilities:

|        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|
| 0.7351 | 0.2321 | 0.0305 | 0.0021 | 0.0001 | 0.0000 | 0.0000 |
| 0.0387 | 0.7453 | 0.1945 | 0.0204 | 0.0011 | 0.0000 | 0.0000 |
| 0.0020 | 0.0778 | 0.7514 | 0.1560 | 0.0123 | 0.0004 | 0.0000 |
| 0.0001 | 0.0061 | 0.1170 | 0.7535 | 0.1170 | 0.0061 | 0.0001 |
| 0.0000 | 0.0004 | 0.0123 | 0.1560 | 0.7514 | 0.0778 | 0.0020 |
| 0.0000 | 0.0000 | 0.0011 | 0.0204 | 0.1945 | 0.7453 | 0.0387 |
| 0.0000 | 0.0000 | 0.0001 | 0.0021 | 0.0305 | 0.2321 | 0.7351 |

File exists and skipping equilibrium rate calculation...  
Loading file instead...

Equilibrium  $r^* = -0.044232$   
File exists and skipping hugget model calculation...  
Loading file instead...  
Loaded hugget model  
States at lower bound ( $a_{\min}=-1.3046$ ): 118  
States at upper bound ( $a_{\max}=65.2287$ ): 0

PS4\_run\_01-eps-converted-to.pdf

PS4\_run\_02-eps-converted-to.pdf

PS4\_run\_03-eps-converted-to.pdf

## Question 2: Aiyagari

```
run('programs/PS4_aiyagari.m'); % run Aiyagari
```

Aggregate labour supply  $L = 1.110326$

K bracket:  $[2.3839, 16434.7938]$

r bracket:  $[0.1208, -0.0992]$

Loading cached K\*...

$K^* = 2.383926$  ( $r^* = 0.120761$ )

Solving full model at  $K^*=2.3839$ ,  $r^*=0.1208$ ,  $w^*=0.8426...$

Loading cached model...

### AIYAGARI EQUILIBRIUM SUMMARY

|       |   |                 |
|-------|---|-----------------|
| $K^*$ | = | 2.3839          |
| $r^*$ | = | 0.1208 (12.08%) |
| $w^*$ | = | 0.8426          |
| $Y^*$ | = | 1.4619          |

```
C*      = 1.2235
K*/Y*   = 1.6307
E[a]    = -0.0000 (should = K* = 2.3839)
```

#### WEALTH DISTRIBUTION STATISTICS

```
Gini coefficient : -0.0000
Top 10% share    : -0.0000 (-0.0%)
Top 1% share     : -0.0000 (-0.0%)
```

#### GRID BINDING CHECK

```
States at lower bound (a_min=0.00): 124 / 1050 (11.8%)
States at upper bound (a_max=60.00): 0 / 1050 (0.0%)
```

All figures saved to figs/



PS4\_run\_04-eps-converted-to.pdf

PS4\_run\_05-eps-converted-to.pdf

PS4\_run\_06-eps-converted-to.pdf

PS4\_run\_07-eps-converted-to.pdf

PS4\_run\_08-eps-converted-to.pdf